

Notes

- increasing numerator, increased the gain

$$\frac{1}{(s+4)} \rightarrow \frac{20}{(s+4)}$$

2

the system has high gain of about 75

The two poles of the system are on the LHP and they are both on the real axis

this gives the time response an over damped curve

the system is stable and has no imaginary oscillatory component ?

Settling time is about 41.5 sec?

3

$$88.6 - 32.6 \rightarrow \text{was } 0.00455 \text{ rad/s} = \omega$$

4. frequency domain

$$\frac{1}{\tau} = \text{time constant}$$

$$= \omega_0$$

= corner

frequency

$$\frac{1}{a} = \text{time constant}$$

the time it takes for e^{-at} to decay to 37% of its initial value

also the time it takes for the step response to rise to 63% of its final value

$$G(s) = \frac{a}{s+a}$$



a is the numerator of the system transfer function

$$\text{time constant} = \frac{1}{a}$$

as a gets bigger, the time constant
like shrinks, meaning it takes shorter time
to get to the 63% of final value

but if a gets smaller t.c. increases, meaning
a smaller numerator of the transfer function with no zeros?
will result in a larger cut off frequency

↓
although that didn't
really work

added positive and negative zeros

increased pole values

increased and decreased gain

U.

added a zero
decrease gain - numerator
move poles further left

S. differences

The poles have changed position and are no longer just on the real axis

the behavior of the signal has changed.

there are more oscillations,

initially it was overdamped, but now it is underdamped with an overshoot

- The gain is significantly lower as well
- Settling time is shorter

Part 2

settling time settle within 3 seconds

percentage overshoot

→ not exceed
5 percent of desired
steady state value,
which is 1 → unit step

kd decreases settling time
as well as percentage overshoot

but I'm guessing
p needs to be higher
even to get the
right output

Z-N rules

- initial controller designs,
- refine with tuning

testing k_p

was trying to test out different values
of k_p